

Electronics build guide

Matek F405-WING V2 / ArduPlane / FlySky / five servos with nose steering

A bench-to-airframe sequence for the supplied MQ9-style model. It defines a wiring plan and acceptance checks. No aircraft was connected, calibrated or flown while preparing this guide.

Remove the propeller for wiring, firmware, calibration, servo and failsafe work. Refit it only for a separate controlled propulsion test after bench checks pass.

Equipment	Build assumption
Flight controller	Matek F405-WING V2; not V1, WTE or TE
Navigation	Matek M9 GPS; exact SKU and compass presence unconfirmed
Controls	Two aileron + two V-tail + one nose-steering servo for supplied gear; 9 g metal gear, model unknown
Power	GenX 4S 5200 mAh LiPo; ReadytoSky 40 A 2-4S ESC advertised as Opto with 5 V / 3 A BEC
Propulsion	Emax ECO II 2807 1300 KV + Gemfan 9045 CL three-blade; combination unqualified
Radio / airspeed	FlySky FS-i6X + FS-IA10B; Matek airspeed SKU unconfirmed

Work in this order

Identify parts -> build and meter harnesses -> configure firmware -> connect radio and sensors -> centre and test servos -> test failsafes -> install and balance -> qualify propulsion. Save a parameter file after each successful stage.

This guide does not assign a flight CG, control throw, stall speed, return altitude or flight gain. Those require the completed airframe, aerodynamic review and measured tests. The supplied CAD includes nose steering (A39); fit five servos for that build. Flaps and payload servos are outside this plan.

Resolve these labels first

Connector shape and wire colour are not a pinout

Unknown	Safe branch before connection
M9 GPS SKU	Read PCB and module labels and obtain that Matek manual. UART module: TX to FC RX3, RX to TX3; match specified supply and ground. Add SDA/SCL only if an I2C compass exists. CAN-only module: stop; this FC has no native CAN. Never infer pin order from another M9.
Airspeed SKU	ASPD-4525: I2C, ARSPD_TYPE=1. ASPD-DLVR: labelled I2C connector, ARSPD_TYPE=9; its CAN connector is different. Verified I2C branch: DA2/SDA + CL2/SCL, ARSPD_BUS=1. Other SKU: use its manual and leave disconnected until identified. [1,6,7]
ESC signal plug	Determine whether red is BEC output, absent, or required optocoupler supply. If BEC output, depin and insulate individually; retain signal and signal ground. If input-side power is required, follow the exact manual. Do not guess from Opto in a listing.
9 g servo model	Confirm voltage, pulse range, current and torque. Use 5 V only if supported. Do not select 6/7.2 V from size or metal gears alone. Five servos cover the supplied controls and steering; quantity is not torque approval.
Battery / connectors	Record mass, dimensions, connector polarity and charge limits. Meter every adapter. Never connect the 4S main lead to receiver, GPS, servo rail or 5 V pad.

Make a connector record

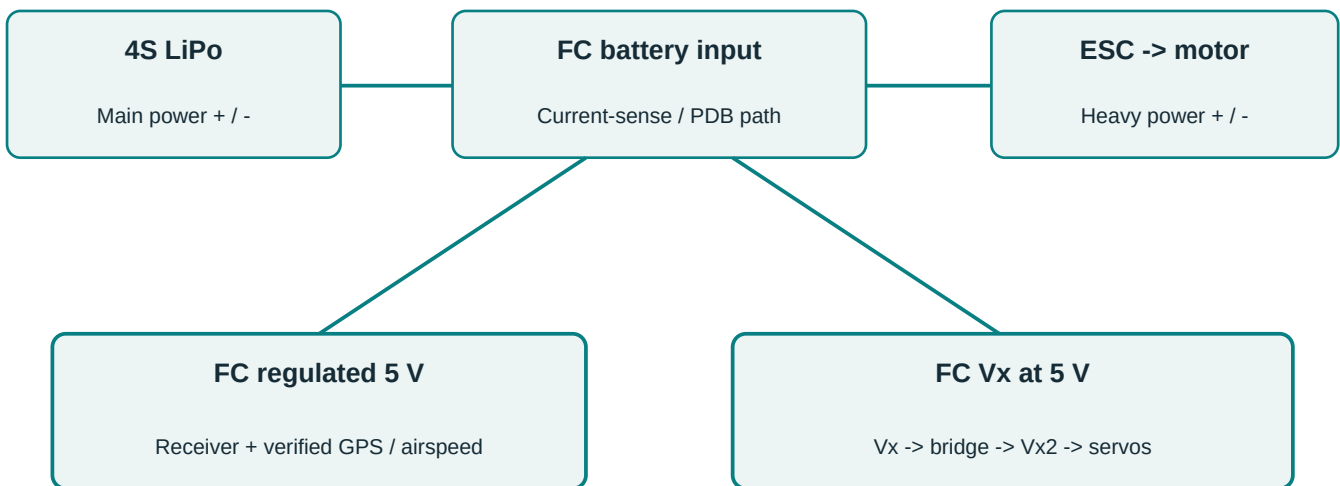
Sketch each plug viewed from a stated face; number cavities; write signal names; continuity-check both ends with all power removed. Label the completed harness. A keyed plug can still be wired incorrectly.

The remaining pages name functions and board pads. They intentionally do not assign cavity numbers for unidentified hardware. Complete this identification before soldering the peripheral harness.

Power architecture

One positive source per rail; common signal ground

Route propulsion current through the FC power-distribution path. This is a functional diagram, not a connector drawing. Match battery-input and ESC-output pads against the V2 layout [1].



ESC signal + ground -> FC S1 + G

ESC BEC output (if present) -> insulated; never joined to FC rails

Build and meter

1. Fit strain-relieved battery and ESC power leads to the documented pads. Keep high-current loops short. Select connectors and wire for measured load; insulate exposed joints.
2. Keep electronics 5 V and servo Vx positives separate. S5-S9 use Vx2: bridge Vx to Vx2 for this internal-BEC plan. Check before plugging in servos. [1]
3. If tests require an external BEC: remove the bridge, verify isolation, power Vx2 from that BEC and move all servo positives to that rail. Retain grounds; never join two BEC positives.
4. With peripherals unplugged, check polarity and hard shorts. First power through a suitable current limiter, then meter 5 V and Vx/Vx2 at far connectors. A limiter may brown out under servo load; it does not qualify flight power.

Wire the signal harness

FC outputs and transmitter channels are independent

FC connection	Destination / setting
S1 + G	ESC PWM signal + ground. SERVO1_FUNCTION=70 (Throttle). Isolate BEC output lead if present.
S2	Unused, SERVO2_FUNCTION=0; leave empty.
S3 + Vx + G	Left aileron. SERVO3_FUNCTION=4 (Aileron).
S4 + Vx + G	Right aileron. SERVO4_FUNCTION=4 (Aileron).
S5 + Vx2 + G	Left tail. Initially SERVO5_FUNCTION=79 (V-tail left). Validate page 8; physical motion governs.
S6 + Vx2 + G	Right tail. Initially SERVO6_FUNCTION=80 (V-tail right). Validate page 8.
S7 + Vx2 + G	Nose steering A39: SERVO7_FUNCTION=26 (GroundSteering). Omit only with a deliberately fixed nosewheel.
S8-S9 / LED	Unused; disconnect. Check SERVO8-10_FUNCTION for unwanted assignments.
RX2 + 5 V + G	FS-iA10B iBUS servo-output signal + supply + ground. Not SENS telemetry port. BRD_ALT_CONFIG=0.
RX3 / TX3 / supply / G	Verified UART GPS: module TX -> RX3, RX -> TX3; supply per exact SKU. SERIAL3_PROTOCOL=5 (GPS).
DA2 / CL2 / supply / G	Verified I2C compass / airspeed: SDA -> DA2, SCL -> CL2. Match pinout and supply. Check shared-bus address conflicts.

This custom allocation replaces standard Plane outputs 1/2/3/4 (aileron/elevator/throttle/rudder). An S1 label alone does not mean throttle. Write and read back each assignment before attaching the ESC. The hardware definition maps S1-S9 to PWM1-PWM9. [2,3,4]

Use conventional PWM initially at 50 Hz (SERVO_RATE), only if supported by identified servos and ESC. Keep DShot disabled for this unknown ESC. Shared timer groups constrain rate/protocol changes. [2]

Firmware and first connection

Mission Planner on Windows; menu names can change

Install the matching target

1. Photograph the board and confirm F405-WING V2. Obtain Mission Planner from ArduPilot. Use a data-capable USB cable with battery, ESC and servos disconnected.
2. Choose stable **ArduPlane**, target **MatekF405-Wing**, version 4.4 or newer. Do not select Copter, INAV, F405-STD or F405-TE. V2 requires newer sensor support. [1]
3. With an ArduPilot bootloader, use SETUP > Install Firmware or Load custom firmware for the matching .apj. Without one, follow the board first-install/DFU procedure using matching firmware containing a bootloader. A missing serial port does not prove the board is dead. [2]
4. Connect; record firmware version, target and vehicle type. Save initial full parameters. Review sensor/storage messages. Insert a supported microSD for logs with power off.

Configure before powering outputs

Use CONFIG > Full Parameter Tree/List for page 4 settings. Write, reboot, reconnect and read back. Use SETUP > Mandatory Hardware for accelerometer, compass, radio and modes; Servo Output for output settings. Parameter names govern if menu wording differs.

Do not bulk-import an internet parameter file. Keep arming checks enabled. Resolve each failed check instead of setting ARMING_CHECK=0. Save the parameters actually read from this board.

Recovery route

Disconnect propulsion power, reconnect USB and compare with the last saved parameter file. Revert the changed settings. After firmware changes, repeat output, sensor and failsafe verification. For a non-booting board, use its documented bootloader recovery with the exact target.

Nose steering: centre A39 with SERVO7_TRIM, limit endpoints mechanically, then set reversal so MANUAL right rudder steers right. Check by pushing the model with motor off. GroundSteering=26 recentres above GROUND_STEER_ALT. For stabilized ground steering, establish a healthy compass, set the altitude gate deliberately, and measure turning diameter for STEER2SRV_P; tune using the official guide before powered taxiing. [14]

Radio input and modes

Plain controls from FS-i6X; mixing inside ArduPlane

Create a new airplane model, select AFHDS 2A and bind the FS-iA10B per its instructions. Disable transmitter V-tail/elevon and custom mixes. Centre trims. Confirm iBUS on the receiver servo serial port. FS-iA10B accepts regulated 5 V; its specification allows 3.5-9 V. [5]

Input	Intended function
CH1	Roll; RCMAP_ROLL=1
CH2	Pitch; RCMAP_PITCH=2
CH3	Throttle; RCMAP_THROTTLE=3
CH4	Yaw; RCMAP_YAW=4
CH5	Three-position mode switch; FLTMODE_CH=5. Measure its actual detent values.
CH6 optional	Dedicated arm/disarm only after verifying the installed firmware option and behaviour; never share with mode control.

Calibrate

Power RX from verified FC 5 V; USB alone may not power it. In Radio Calibration, check that each stick moves only its intended bar. Move sticks and assigned switches through full range; finish throttle low, other sticks centred. Save. [8]

Check RCn_REVERSED semantics: increasing normalized input means roll right, pitch up, more throttle and yaw right. Correct inputs first; physical surface direction is adjusted later using SERVOx_REVERSED. Recalibrate after transmitter endpoint changes. [8]

In Flight Modes, set all six slots MANUAL initially; assign the slot reached by the middle detent to FBWA. Keep the third detent MANUAL until navigation is validated; deliberately assign RTL later. Verify HUD mode at all detents and transition bands.

Receiver voltage telemetry is not automatically propulsion-pack voltage. Separate receiver antennas from carbon, battery and motor wiring, and position active ends per the radio manual. Perform the documented range test after final installation.

Sensors and battery readings

Calibrate in final orientation; inspect the actual response

Attitude, heading and position

Mount FC securely and set AHRS_ORIENTATION to match. Run six-position accelerometer calibration, then level calibration on an accurately level aircraft datum. Roll and pitch the aircraft; verify HUD attitude follows correctly before stabilized-mode tests.

Determine whether GPS includes a compass; FC has none onboard. Set external compass orientation and calibrate away from steel, magnets and power wires. Point the aircraft in several known directions and compare heading independently. Repeat after final installation. [2]

For verified UART M9, use SERIAL3_PROTOCOL=5 and an appropriate GPS type (usually Auto for a u-blox UART module). Acquire outdoors with clear sky. Verify plausible position, fix and time. A power LED is not navigation health; GPS-only hardware does not provide magnetic heading.

Battery monitor - V2 values

Parameter	Initial value
BATT_MONITOR / voltage / current pins	4 / BATT_VOLT_PIN=10 / BATT_CURR_PIN=11
BATT_VOLT_MULT / BATT_AMP_PERVLT	11.0 / 66.7; older-board 31.7 is wrong for V2
BATT_CAPACITY	5200 mAh nominal; validate usable capacity separately

Compare reported voltage with a reliable meter simultaneously. Correction: new multiplier = old multiplier x meter voltage / reported voltage. Compare current with a calibrated inline instrument at steady meaningful load; new current scale = old scale x reference current / reported current. Do not fit scale near zero. Check idle offset and consumed capacity through a measured discharge. [1,4]

Choose low/critical thresholds, reserve, delays and actions from pack specification and loaded behaviour, then record on page 11. C-rating does not prove motor current, usable capacity or endurance.

Servos and V-tail direction

Left/right viewed from behind the aircraft, facing forward

Centre each servo electrically before fitting its horn. Align linkages mechanically, then use SERVOx_TRIM for small corrections. Establish conservative endpoints and inspect every hinge and pushrod through travel. No generic travel angle is assigned.

Test	Required response
MANUAL: roll right	Right aileron trailing edge up, left down; reverse for roll left.
MANUAL: pitch up	Both tail trailing edges move in upward pitch-producing sense; reverse for pitch down.
MANUAL: yaw right	Both tail trailing edges move toward aircraft right; reverse for yaw left.
FBWA, centred sticks: tilt aircraft right	Left aileron up, right down: correction opposes tilt.
FBWA: raise nose	Tail command produces nose-down correction; lowering nose produces nose-up correction.

CAD inspection shows an upright V-tail. Initial 79/80 assignments still require physical direction checks. Per channel: wrong in both pitch and yaw -> reverse output; correct for one axis only -> exchange function 79/80 and repeat. Final function labels may not match physical side. [9]

Use the V-tail roll-to-rudder check only with nonzero KFF_RDDRMIX and restore its intended value afterward. A stationary yaw-rotation test is not a heading-hold test. [9]

Combined commands and supply loading

Apply simultaneous full pitch/yaw in all four combinations. Check binding, over-centre horns, pushrod buckling, buzz and saturation. Adjust mechanics, endpoints and MIXING_GAIN together. Repeat MANUAL and FBWA tests after each change. [9]

Exercise all installed servos together under representative load without prolonged stall. Capture Vx and 5 V minima, current peaks, temperature and resets. A multimeter can miss brief dips. Vx is rated 5 A continuous / 6 A peak; electronics 5 V is 2 A. These are ceilings, not measured margin. [1]

Airspeed and propulsion

Sensor response is not calibration; motor spin is not thrust

Airspeed branch

After SKU verification, connect page 2 I2C harness, set ARSPD_TYPE and ARSPD_BUS, reboot and check health. ASPD-4525 uses type 1; DLVR-L10D I2C uses type 9. Never connect the DLVR CAN socket to I2C. [6,7]

Put pitot in clean airflow away from prop wash. Connect dynamic/static lines per probe manual, without kinks or leaks. Allow ambient-temperature settling. Loosely shield pitot from wind without sealing ports; run Preflight Calibration for zero. [10]

Use gentle airflow toward probe and verify a positive response returning near zero. Avoid strong pressure, liquid or blowing into tubing. This proves response only. Keep ARSPD_USE=0 during initial validation, but some current Plane versions still use a healthy enabled sensor for speed scaling; this is not complete isolation. [10,11]

If suspect, disable sensor (ARSPD_TYPE=0) and resolve before flight planning. Ratio calibration requires an independently safe aircraft and controlled flight/log review. A bench airflow check cannot establish true airspeed or stall speed.

ESC and motor

Mounting: the Emax drawing calls out four M3 holes on a 19 mm pitch circle (13.435 mm adjacent square spacing), despite marketing saying 19x19. Check the actual motor with a paper template before drilling. Select screw length from mount thickness and measured thread depth; screws must not reach windings. Emax lists 6-7 inch props; the supplied 9-inch combination needs measured qualification. [15]

Confirm ESC manual and PWM range. High/low calibration requires FC and receiver to remain powered while ESC power is separately switched. USB may not power this RX rail: use a documented isolated setup or separate servo tester; do not backfeed a rail. [12]

Only if confirmed by ESC manual: prop removed, ESC unpowered, command maximum; power ESC, await specified calibration indication, command minimum, verify acceptance. Otherwise follow its documented method. Mission Planner multicopter motor calibration is not for Plane. [12]

Verify motor stopped when disarmed and at minimum. Briefly check low-speed rotation with airframe restrained. For conventional brushless reversal, power off and swap two phase wires. Prop handedness and orientation must produce forward thrust in the pusher installation.

Failsafes: prove the trigger

Prop removed; disconnect motor power for configuration tests

A switch labelled failsafe is not a link-loss test. With Mission Planner recording, power the assembled radio path, select a test mode and switch the transmitter off. Observe inputs, failsafe messages and resulting mode; restore link and observe recovery.

Radio loss

Enable `THR_FAILSAFE=1`. Determine whether RX stops frames, flags loss or continues sending preset/held channels. Held normal values may prevent detection. Configure a detectable loss condition and repeat the actual transmitter-off test. [13]

For low-throttle triggering, place `THR_FS_VALUE` between measured loss throttle and lowest valid throttle, with reliable separation. Example only: valid low 1000, loss 900, threshold 950 microseconds. If RX cannot provide a distinct condition, change the setup.

Candidate fixed-wing policy: short loss `CIRCLE` (`FS_SHORT_ACTN=1`), sustained loss `RTL` (`FS_LONG_ACTN=1`). Record both timeouts and verify firmware action labels. Home, return height, terrain clearance and available power must be valid before using this flight policy. [13]

Battery and navigation failures

Record `BATT_LOW_VOLT`, `BATT_CRT_VOLT`, `BATT_LOW_MAH`, `BATT_CRT_MAH`, `BATT_LOW_TIMER` and low/critical actions. Confirm units and action labels in installed firmware. Test using a suitable variable supply within FC limits or temporary threshold changes; restore, read back and save flight values afterward. Do not deeply discharge LiPo to trigger a test.

Ground RTL testing proves a requested mode change, not a safe return path. Test navigation-failure policies in simulation before controlled flight evaluation with an experienced pilot. Do not unplug live GPS cables or obstruct sensors near a spinning prop.

Record trigger, delay, action and recovery. Repeat from every intended flight mode. If a failsafe cannot be observed or explained, keep the flight gate closed.

Installation and release record

Complete on the actual aircraft

Keep the battery tray adjustable and restrained in all directions. Separate high-current wires from GPS/compass and RX antennas. Strain-relieve connectors and protect wires at foam/wood/carbon edges. Preserve access to USB, SD card, rail bridge and battery connector. Label detachable wing/tail leads.

Provisional setup balance station

Mark 522 mm aft of the CAD nose tip, or 22.2 mm aft of the front wing-joiner axis. This is the geometric quarter-MAC initial balancing reference, not an aerodynamic or flight-approved CG range. CAD mesh integration gives exposed wing area 304,451 mm², MAC 169.827 mm and quarter-MAC station 522.312 mm aft nose. Flight release still needs measured mass, control authority and experienced-pilot review.

xCG = sum(mass x position) / total mass

Battery adjustment: battery shift = total mass / battery mass x desired CG shift.

Use one nose datum, aft positive. Include wiring, glue and finish. Mark repeatable pack positions; verify assembled balance and lateral balance.

Record	Measured / approved value
Firmware / saved parameters	_____
5 V minimum / Vx minimum / servo peak current	_____
Voltage calibration / current reference	_____
RC loss trigger / short / long / recovery	_____
Battery low / critical / actions	_____
Total mass / battery mass / actual CG from nose	_____
Desired CG shift / required battery movement	_____
Control throws / propulsion test / flight CG review	_____

Flight gate: identified connectors; stable loaded supplies; correct control/stabilization direction; range and failsafe checks; valid sensors; structural/CG approval; qualified propulsion. Motor + 9045 three-blade + 40 A ESC compatibility remains unproved. A prop-off spin does not qualify it. Use a proper restrained fixture and experienced supervision; measure current, thrust, voltage and temperature.

References and handoff

Primary sources checked 19 September 2026

[1] Matek V2 specifications, layout and mapping

<https://www.mateksys.com/?portfolio=f405-wing-v2>

[2] ArduPilot F405-Wing (older details need V2 cross-check)

<https://ardupilot.org/plane/docs/common-matekf405-wing.html>

[3] ArduPilot Choosing Servo Functions

<https://ardupilot.org/plane/docs/servo-functions.html>

[4] ArduPilot board hardware definition (master)

https://github.com/ArduPilot/ardupilot/blob/master/libraries/AP_HAL_ChibiOS/hwdef/MatekF405-Wing/hwdef.dat

[5] FlySky FS-iA10B specification

<https://www.flysky-cn.com/ia10b-canshu>

[6] Matek ASPD-4525

<https://www.mateksys.com/?portfolio=aspd-4525>

[7] Matek ASPD-DLVR

<https://www.mateksys.com/?portfolio=aspd-dlvr>

[8] ArduPilot RC input mapping and reversal

<https://ardupilot.org/plane/docs/rc-throw-trim.html>

[9] ArduPilot V-tail / inverted V-tail setup

<https://ardupilot.org/plane/docs/guide-vtail-plane.html>

[10] ArduPilot Airspeed Calibration

<https://ardupilot.org/plane/docs/calibrating-an-airspeed-sensor.html>

[11] ArduPilot Airspeed Parameters Setup

<https://ardupilot.org/plane/docs/airspeed-parameters-setup.html>

[12] ArduPilot ESC Calibration

<https://ardupilot.org/plane/docs/common-esc-calibration.html>

[13] ArduPilot Plane Failsafe Function

<https://ardupilot.org/plane/docs/apms-failsafe-function.html>

[14] ArduPilot Ground Steering

<https://ardupilot.org/plane/docs/tuning-ground-steering-for-a-plane.html>

[15] Emax ECO II 2807 official drawing in product description

<https://emaxmodel.com/products/emax-eco-ii-series-2807-1300kv-1700kv-1500kv-brushless-motor-for-rc-drone-fpv-racing>

Continuity: output/pdf/MQ9_Hardware_and_CG_Notes.pdf, 19 September 2026. Exact ESC, servo, GPS and airspeed manuals remain required. No aircraft current, thrust, servo load, firmware readback or flight evidence was collected.

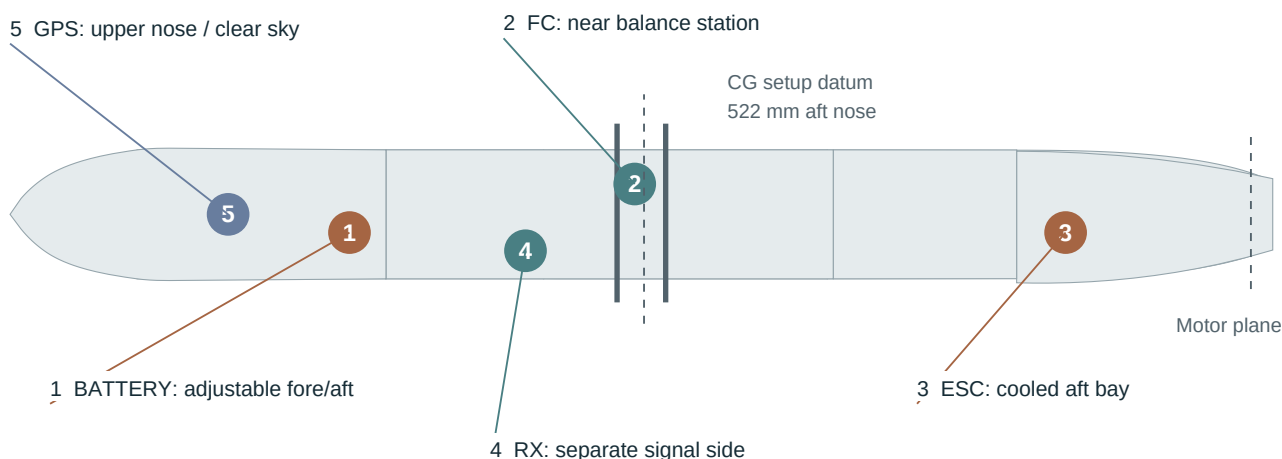
Where the electronics go

Proposed zones on the actual CAD body - not mounting-hole templates

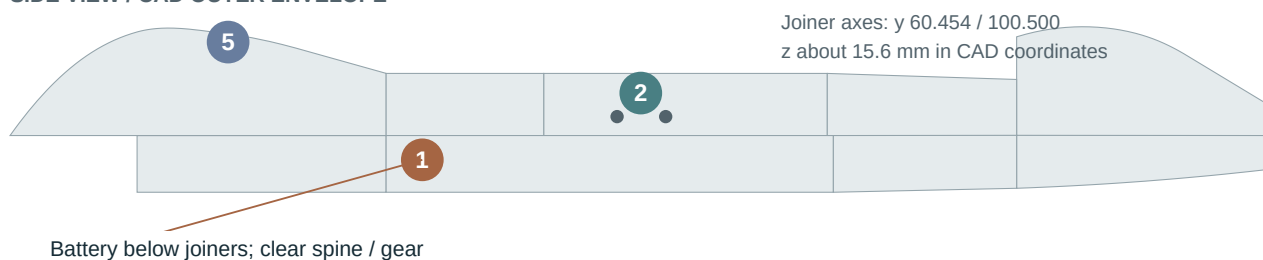
Place equipment with the bare structure open. These zones follow the assembled fuselage and wing-joiner geometry; they do not confirm battery fit. The GenX pack dimensions and its cable exit must be checked before a tray or hatch is made.

TOP VIEW / NOSE LEFT

PUSHER AFT



SIDE VIEW / CAD OUTER ENVELOPE

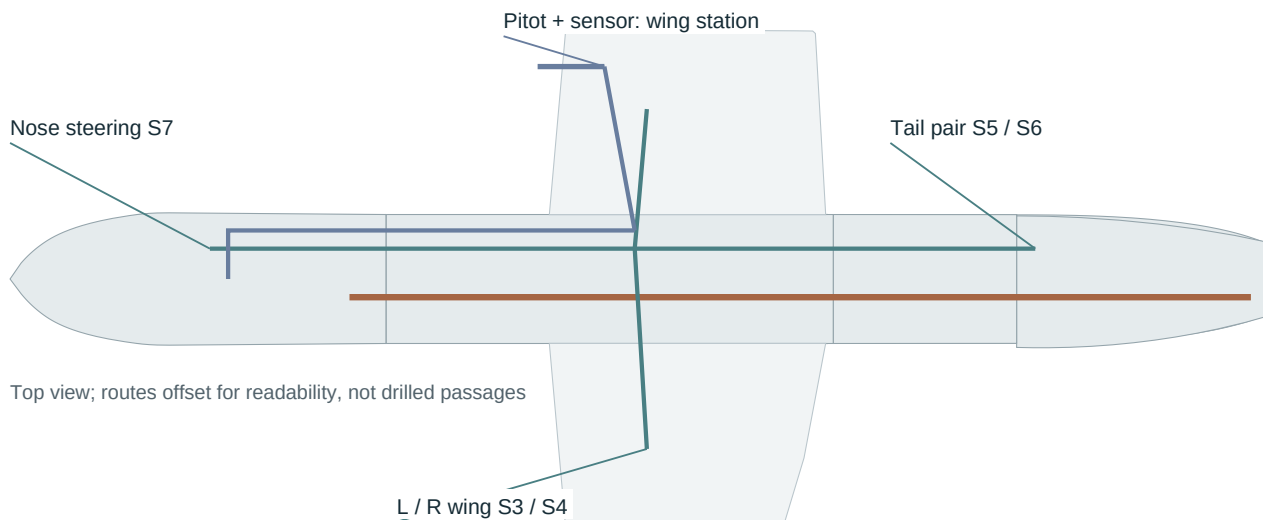


Zone	Mounting decision
1 Battery	Adjustable tray below the wing tubes, to one side of the centre wood spine as space permits. Clear nose/main gear and tube roof. Positive restraint in every direction; accessible connector. No pack dimensions or fitted tray claimed.
2 Flight controller	Accessible near the setup balance station on a firm mounting platform, using documented board grommets. Arrow forward; clear USB/SD access, level datum and wires that pull on the board.
3 / 4 / 5 ESC, RX, GPS	ESC aft with airflow and service access; RX on the signal side. Put M9 antenna upward with clear sky; if compass equipped, separate it from battery/current wiring and steel. Place RX active antenna ends clear of carbon and motor power. [16]

Wing pitot placement and harness routes are shown next. Colour marks identify functions; their drawn size and coordinates are not component footprints.

Harness routes and mounting

Keep removable joints accessible; confirm fit before closing skins



Harness	Route and service point
Brown: high current	Battery -> FC distribution -> ESC -> motor. Keep + and - together and away from signal/sensor bundles. Keep ESC input leads short; obtain the actual ESC maker cable-length/capacitor guidance before extending. No unsupported length limit is assumed.
Teal: control	Separate 3-wire S3/S4 wing leads, labelled L / R with signal, servo positive and ground. Retain keyed connectors at wing roots. Route S5/S6 to the tail and S7 to nose steering; secure leads clear of hinges, pushrods and gear.
Blue: navigation	GPS UART and optional compass to FC; airspeed I2C to a supported sensor near its pitot. Mount probe upstream of the wing leading edge in undisturbed flow, away from fuselage boundary layer and prop wash. Keep tubes short, unkinked and removable. Do not choose a fuselage-roof position just because it is convenient. [17]

Install in six passes

1. Mock up actual battery, FC, ESC and receiver; mark hatch access. 2. Confirm tray travel around spine, gear and joiners. 3. Install mounts, then route power on one side and signal wiring separately. 4. Add labelled wing/tail service connectors with strain relief. 5. Check pack restraint, cooling, full linkage travel and antenna exposure with hatches shut. 6. Repeat rail-load, radio-range, sensor, control-direction and balance checks before closing the build.

CAD basis: assembled meshes A06/A08/A12/A15/A17/A20/A42/A56/A59; y increases aft. Nose y -439.65, motor mounting plane y 582.65, quarter-MAC setup station y 82.662 mm. Zones are engineering placement proposals, not proven fits.

[16] ArduPilot GPS/compass mounting

[17] ArduPilot Pitot Tube Considerations